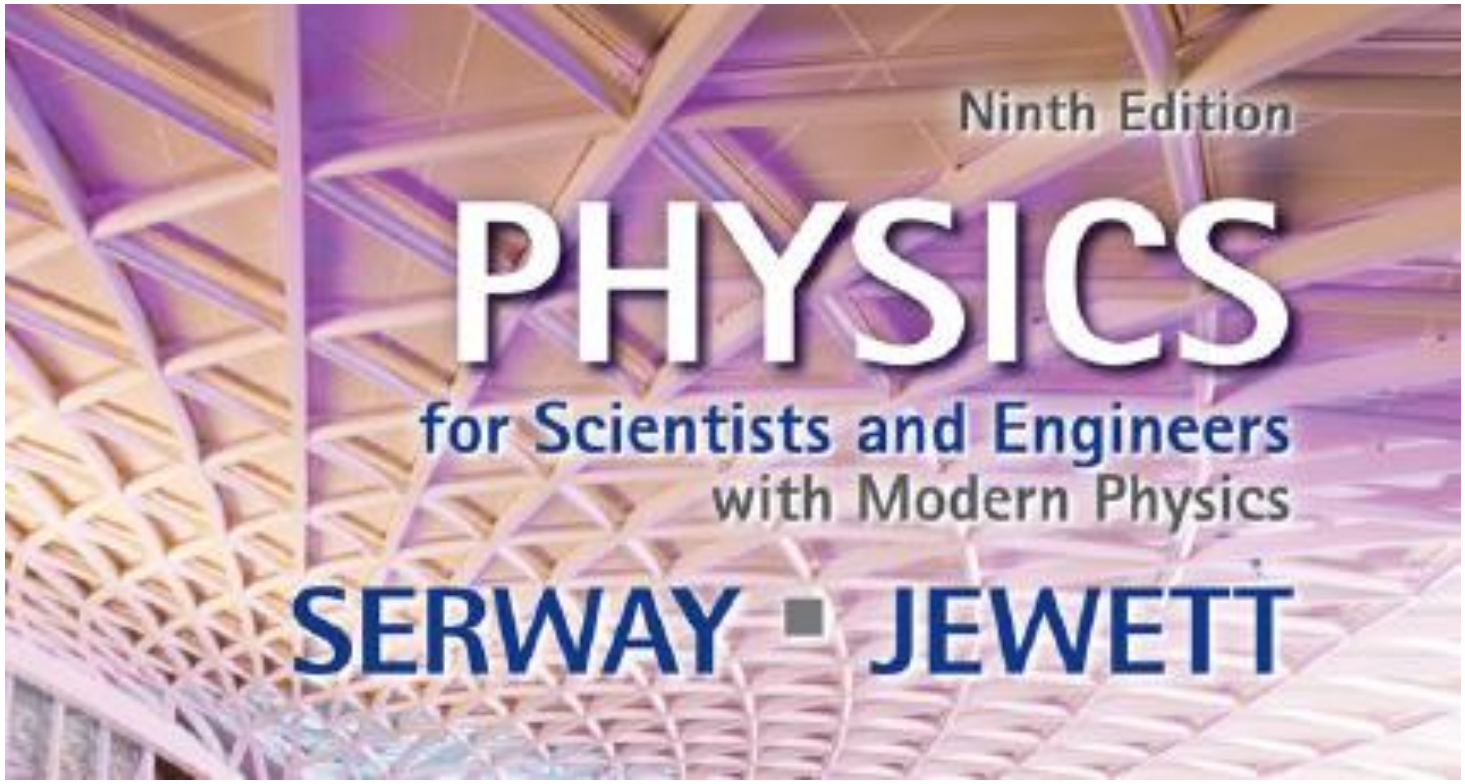
Physics I


Ninth Edition
PHYSICS
for Scientists and Engineers
with Modern Physics
SERWAY ■ JEWETT

Textbook

- Physics for scientists and engineers

Authors: Serway and Jewetts

9th edition

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الاستاذ الدكتور زياد عليمات

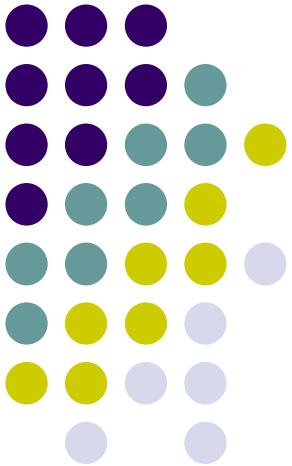
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Chapter 1

Chapter 1: Physics and Measurement

(Introduction)



Definition of Physics:



- **Physics, the most fundamental physical science, is concerned with the fundamental principles and natural phenomena of our Universe.**
- **Physics put a physical laws to understand these phenomena that occurs in our universe.**
- **To describe natural phenomena, we must make measurements of various aspects of nature. Each measurement is associated with a physical quantity, such as the length of an object.**
- **These physical laws consist of physical quantities like (force, velocity, acceleration, displacement, time, …..).**
- **Physical laws are expressed by mathematical notations, for example (Newton's second law of motion: F =ma): F (force), m (mass), a (acceleration)**

Objectives of Physics

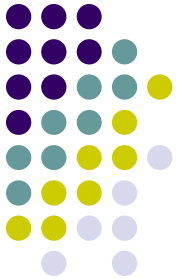
- To find the limited number of fundamental laws that govern natural phenomena
- To use these laws to develop theories that can predict the results of future experiments
- Express the laws in the language of mathematics
  - Mathematics provides the bridge between theory and experiment
  - Measurements used to describe natural phenomena

Physical Quantities:

In mechanics the physical quantities divided to Basics and Derived quantities.

- The basic quantities in mechanics are three: length, mass, and time.
- The derived quantities: All other quantities in mechanics can be expressed in terms of these three. For examples:
- Speed = distance / time =length /time (note: speed is a scalar quantity)
- Velocity = displacement / time =length /time (note: velocity is a vector quantity)
- Force = mass*acceleration= mass*(velocity/time) = mass*(length/time^2)

Definition of length, mass, and time



- **Length:** We can identify **length as the distance between two points in space.**
  - The SI fundamental unit of length is meter that defined as **the distance traveled by light in vacuum during a time of 1/299 792 458 second. In effect, this latest definition establishes that** the speed of light in vacuum is precisely **299 792 458 meters per second.**
- **Mass:** The SI fundamental unit of **mass, the kilogram (kg), is defined as the mass of a specific platinum–iridium alloy cylinder kept at the International Bureau of Weights and Measures at Sèvres, France.**
- **Time** The SI fundamental unit of **time, is second. One second is now defined as 9 192 631 770 times the period of vibration of radiation from the cesium-133 atom.**

SI Units: SI – Systéme International

Fundamental Quantities and Their Units

mks-System

Quantity	SI Unit
Length	meter
Mass	kilogram
Time	second

SI Units: SI – Systéme International

Fundamental Quantities and Their Units

cgs-System

Quantity	SI Unit
Length	centimeter
Mass	gram
Time	second

SI Units: SI – Systéme International

Fundamental Quantities and Their Units

US Customary System

- Still used in the US, but text will use SI

Quantity	Unit
Length	foot
Mass	Slug (14.5939 kg)
Time	second

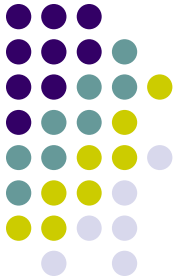
Prefixes

- Prefixes correspond to powers of 10
- Each prefix has a specific name
- Each prefix has a specific abbreviation

Table 1.4 Prefixes for Powers of Ten

Power	Prefix	Abbreviation	Power	Prefix	Abbreviation
10^{-24}	yocto	y	10^{3}	kilo	k
10^{-21}	zepto	z	10^{6}	mega	M
10^{-18}	atto	a	10^{9}	giga	G
10^{-15}	femto	f	10^{12}	tera	T
10^{-12}	pico	p	10^{15}	peta	P
10^{-9}	nano	n	10^{18}	exa	E
10^{-6}	micro	μ	10^{21}	zetta	Z
10^{-3}	milli	m	10^{24}	yotta	Y
10^{-2}	centi	c			
10^{-1}	deci	d			

Basic Quantities and Their Dimension



- Dimension has a specific meaning – it denotes the physical nature of a quantity
- Dimensions are denoted with square brackets:
  - Length [L]
  - Mass [M]
  - Time [T]

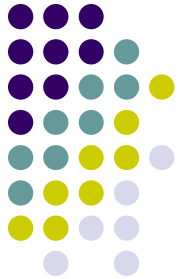
Dimensions and Units

- Each dimension can have many actual units
- Table 1.5 for the dimensions and units of some derived quantities

Table 1.5 **Dimensions and Units of Four Derived Quantities**

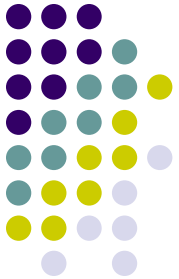
Quantity	Area (A)	Volume (V)	Speed (v)	Acceleration (a)
Dimensions	L^2	L^3	L/T	L/T^2
SI units	m^2	m^3	m/s	m/s^2
U.S. customary units	ft^2	ft^3	ft/s	ft/s^2

Dimensional Analysis



- Technique to check the correctness of an equation or to assist in deriving an equation
- Both sides of equation must have the same dimensions
- Any relationship can be correct only if the dimensions on both sides of the equation are the same
- Cannot give numerical factors: this is its limitation

Dimensional Analysis, example



- Given the equation: $x = \frac{1}{2} at^2$
- Check dimensions on each side:

$$L = \frac{L}{T^2} \cdot T^2 = L$$

- The T^2's cancel, leaving L for the dimensions of each side
  - The equation is dimensionally correct
  - There are no dimensions for the constant

Dimensional Analysis, example

Example 1.1 Analysis of an Equation

Show that the expression $v = at$, where v represents speed, a acceleration, and t an instant of time, is dimensionally correct.

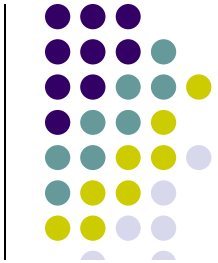
SOLUTION

Identify the dimensions of v from Table 1.5: $[v] = \frac{L}{T}$

Identify the dimensions of a from Table 1.5 and multiply by the dimensions of t: $[at] = \frac{L}{T^2} T = \frac{L}{T}$

Therefore, $v = at$ is dimensionally correct because we have the same dimensions on both sides. (If the expression were given as $v = at^2$, it would be dimensionally *incorrect*. Try it and see!)

Dimensional Analysis, example



Example 1.2 Analysis of a Power Law

Suppose we are told that the acceleration a of a particle moving with uniform speed v in a circle of radius r is proportional to some power of r, say r^n, and some power of v, say v^m. Determine the values of n and m and write the simplest form of an equation for the acceleration.

SOLUTION

Write an expression for a with a dimensionless constant of proportionality k:

$$a = kr^n v^m$$

Substitute the dimensions of a, r, and v:

$$\frac{\mathrm{L}}{\mathrm{T}^2} = \mathrm{L}^n \left(\frac{\mathrm{L}}{\mathrm{T}}\right)^m = \frac{\mathrm{L}^{n+m}}{\mathrm{T}^m}$$

Equate the exponents of L and T so that the dimensional equation is balanced:

$$n + m = 1 \text{ and } m = 2$$

Solve the two equations for n:

$$n = -1$$

Write the acceleration expression:

$$a = kr^{-1} v^2 = k\frac{v^2}{r}$$

Conversion of Units

$1\ \text{mile} = 1\,609\ \text{m} = 1.609\ \text{km}$ $\quad$ $1\ \text{ft} = 0.304\,8\ \text{m} = 30.48\ \text{cm}$

$1\ \text{m} = 39.37\ \text{in.} = 3.281\ \text{ft}$ $\quad$ $1\ \text{in.} = 0.025\,4\ \text{m} = 2.54\ \text{cm}\ \text{(exactly)}$

For example, suppose we wish to convert 15.0 in. to centimeters. Because 1 in. is defined as exactly 2.54 cm, we find that

$$15.0\ \text{in.} = (15.0\ \cancel{\text{in.}}\,)\left(\frac{2.54\ \text{cm}}{1\ \cancel{\text{in.}}}\right) = 38.1\ \text{cm}$$

Problems with solutions

9. Which of the following equations are dimensionally correct? (a) $v_f = v_i + ax$ (b) $y = (2\text{ m})\cos(kx)$, where $k = 2\text{ m}^{-1}$

(a) Write out dimensions for each quantity in the equation

$$v_f = v_i + ax$$

The variables v_f and v_i are expressed in units of m/s, so

$$[v_f] = [v_i] = LT^{-1}$$

The variable a is expressed in units of m/s^2; $[a] = LT^{-2}$

The variable x is expressed in meters. Therefore, $[ax] = L^2T^{-2}$

Consider the right-hand member (RHM) of equation (a):

$$[RHM] = LT^{-1} + L^2T^{-2}$$

Quantities to be added must have the same dimensions.

Therefore, equation (a) **is not** dimensionally correct.

(b) Write out dimensions for each quantity in the equation

$$y = (2\text{ m})\cos(kx)$$

For y, $[y] = L$

for 2 m, $[2\text{ m}] = L$

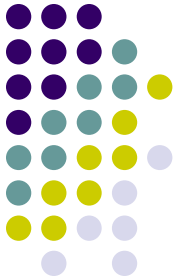
and for (kx), $[kx] = \left[\left(2\text{ m}^{-1}\right)x\right] = L^{-1}L$

Therefore we can think of the quantity kx as an angle in radians, and we can take its cosine. The cosine itself will be a pure number with no dimensions. For the left-hand member (LHM) and the right-hand member (RHM) of the equation we have

$[LHM] = [y] = L$ $[RHM] = [2\text{ m}][\cos(kx)] = L$

These are the same, so equation (b) **is** dimensionally correct.

14. (a) Assume the equation $x = At^3 + Bt$ describes the motion of a particular object, with x having the dimension of length and t having the dimension of time. Determine the dimensions of the constants A and B. (b) Determine the dimensions of the derivative $dx/dt = 3At^2 + B$.



Summed terms must have the same dimensions.

(a) $[X] = [At^3] + [Bt]$

$L = [A]T^3 + [B]T \rightarrow \boxed{[A] = L/T^3, \text{ and } [B] = L/T}.$

(b) $[dx/dt] = [3At^2] + [B] = \boxed{L/T}.$

15. **W** A solid piece of lead has a mass of 23.94 g and a volume of 2.10 cm^3. From these data, calculate the density of lead in SI units (kilograms per cubic meter).

$$\rho = \left(\frac{23.94\ \text{g}}{2.10\ \text{cm}^3}\right)\left(\frac{1\ \text{kg}}{1\ 000\ \text{g}}\right)\left(\frac{100\ \text{cm}}{1\ \text{m}}\right)^3$$

$$= \left(\frac{23.94\ \text{g}}{2.10\ \text{cm}^3}\right)\left(\frac{1\ \text{kg}}{1\ 000\ \text{g}}\right)\left(\frac{1\ 000\ 000\ \text{cm}^3}{1\ \text{m}^3}\right)$$

$$= \boxed{1.14 \times 10^4\ \text{kg/m}^3}$$

17. A rectangular building lot has a width of 75.0 ft and a length of 125 ft. Determine the area of this lot in square meters.

For a rectangle, Area = Length × Width. We use the conversion 1 m = 3.281 ft. The area of the lot is then

$$A = LW = (75.0\ \text{ft})\left(\frac{1\ \text{m}}{3.281\ \text{ft}}\right)(125\ \text{ft})\left(\frac{1\ \text{m}}{3.281\ \text{ft}}\right) = \boxed{871\ \text{m}^2}$$

24. **M** A house is 50.0 ft long and 26 ft wide and has 8.0-ft-high ceilings. What is the volume of the interior of the house in cubic meters and in cubic centimeters?

The volume of the interior of the house is the product of its length, width, and height. We use the conversion 1 ft = 0.304 8 m and 100 cm = 1 m.

$$V = LWH$$

$$= (50.0\ \text{ft})\left(\frac{0.304\ 8\ \text{m}}{1\ \text{ft}}\right) \times (26\ \text{ft})\left(\frac{0.304\ 8\ \text{m}}{1\ \text{ft}}\right) \times (8.0\ \text{ft})\left(\frac{0.304\ 8\ \text{m}}{1\ \text{ft}}\right)$$

$$= 294.5\ \text{m}^3 = \boxed{290\ \text{m}^3}$$

$$= (294.5\ \text{m}^3)\left(\frac{100\ \text{cm}}{1\ \text{m}}\right)^3 = \boxed{2.9 \times 10^8\ \text{cm}^3}$$